ABHINAV MITTRA

Software Engineer 3 | Full Stack and Agentic AI Systems

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Software Engineer with 6+ years at Cisco specialising in full stack architecture, high throughput micro services, and production agentic AI systems. Proven track record in multi agent orchestration, GraphRAG retrieval, LLM evaluation layers, container security, and legacy system modernisation.

80% MTTR Reduction	**10,000+ QPS / <100ms**	**95.2% Factual Grounding**	**50% Load Time Gain**
Automated Incident Queue Agent	High Throughput Integration APIs	Ragas / LLM as a Judge Evals	Struts to MEAN Stack Migration

PROFESSIONAL EXPERIENCE

Cisco Systems | Software Engineer 3 (Nov 2023 - Present) - Bengaluru, India

- **Contributing to the Finance Discovery Workbench using LangGraph, LangChain, and Next.js** to automate financial requirements validation and solution document generation. Integrated GraphRAG (Neo4j / ColBERTv2) and built an automated LLM as a Judge evaluation layer (Ragas/TruLens), reducing analysis drafting time by 75% (from 16h to 4h) while maintaining <0.8% hallucination rates across 500+ enterprise schemas.
- **Engineered a dynamic fault tolerant Incident Resolution Agent** utilising asynchronous state snapshotting (Write Ahead Logging via Redis and PostgreSQL) and secondary feedback learning loops, automated queue handling to slash Mean Time To Resolution (MTTR) by 80% (from 4.5h to 12 mins) across 200+ monthly enterprise tickets.
- **Designed and scaled high concurrency Angular/Node.js micro frontends on OpenShift** backed by event driven MuleSoft and Kafka integration pipelines, processed 5,000+ daily active user workflows with 99.99% service availability and sub 100ms API response times.
- **Hardened cloud-native deployments by migrating 35+ micro service containers to distroless Chainguard images** resolving 140+ container security vulnerabilities and shrinking container image footprints by 62% (from 800MB to 304MB).
- **Led legacy to modern architecture migration of Struts applications to MEAN stack** refactoring monolithic endpoints into modular Node.js services and optimising MongoDB aggregation pipelines, resulting in a 50% lower page load latency (p95 dropped from 3.2s to 1.6s) for 1,000+ reseller partners.

Cisco Systems | Software Engineer 2 (Jun 2021 - Nov 2023) - Bengaluru, India

- **Developed and optimised a REST API inference pipeline for BERT incident classification** exporting models to ONNX Runtime with dynamic INT8 quantisation; reduced model VRAM/RAM footprint by 65% and achieved p99 latency <18ms at 3,200 QPS, elevating incident routing accuracy from 71% to 89%.
- **Architected enterprise MuleSoft REST and gRPC integration services** for reseller management and order booking processing 0M+ in annual bookings; implemented Redis distributed caching and circuit breakers to guarantee zero payload loss during peak traffic surges.

Cisco Systems | Software Engineer (Aug 2020 - Jun 2021) - Bengaluru, India

- **Optimised legacy Spring MVC micro services and database query execution plans** by tuning HikariCP connection pools and JVM garbage collection parameters, achieving a 20% reduction in average API response time (from 450ms to 360ms).
- **Constructed a centralised observability and distributed tracing dashboard using Splunk and ELK Stack** providing real-time telemetry across 12 core integration services and accelerating incident diagnosis time by 35%.

HIGH IMPACT AI AND SYSTEMS PROJECTS

GraphRAG Sub-Graph Extraction & Late-Interaction Retrieval Engine | *Stack: Python 3.12, Qdrant, Neo4j, FastGraphRAG, ColBERTv2, FastAPI, Docker*

- **Designed a hybrid GraphRAG and ColBERTv2 retrieval engine** indexing 500,000+ unstructured corporate documents into a Neo4j knowledge graph and Qdrant multi vector store for multi hop enterprise reasoning.
- **Created an automated sub graph pruning algorithm** using personalised PageRank to isolate top k entity clusters, cutting prompt token usage by 55% while boosting multi hop QA accuracy by 34% (HotpotQA benchmark) and dropping retrieval latency to 210ms (p95).

Autonomous CI/CD Automated Testing & Static Code Quality Verification Agent | *Stack: Python, Rust, Tree-sitter, Semgrep, GitHub Actions API, Docker, LLM Function Calling*

- **Architected an automated PR quality verification agent** integrating Tree sitter AST differential parsing and Semgrep static analysis rules into GitHub Actions pipelines across 10+ active repositories.
- **Deployed an LLM-guided dynamic test runner** executing verification inside isolated Docker sandboxes, suppressing false positives, cutting developer triage time by 62%, and catching 23 critical concurrency and validation risks prior to release.

High Throughput Synthetic Data Engine with Multi-Stage Quality Filtering | *Stack: Python, Ray Data, Polars, DuckDB, MinHash LSH, FastText, Hugging Face Datasets*

- **Engineered a distributed synthetic data pipeline on Ray Data** processing 1,000,000+ synthetic token generations per day across an AWS EKS cluster.
- **Designed a custom C++/Python MinHash LSH deduplication module** filtering out 38% near duplicate passages to save ~2k in pre training compute, and enforced a 3-tier quality filter (FastText + Perplexity bounds + LLM as a Judge) boosting benchmark dataset quality by 22%.

TECHNICAL SKILLS

Frontier AI & LLMOps: LangGraph, LangChain, GraphRAG (Neo4j), ColBERTv2, Ragas, TruLens (LLM Evals), Prompt & Agentic Orchestration, vLLM, ONNX Runtime, Hugging Face

Full Stack & Systems: Python, Node.js, Next.js, Angular, C++, Java / Spring Boot, Rust, REST APIs, gRPC, Flask, Micro-frontends

Data & Integration: MuleSoft, Kafka, Snowflake, Qdrant, Neo4j, Redis, MongoDB, PostgreSQL, Oracle, Polars, DuckDB

Cloud, DevOps & Tooling: OpenShift, Docker, Chainguard (Distroless), Kubernetes, Jenkins, Spinnaker, Splunk, ELK Stack, Semgrep, Tree sitter

EDUCATION

Bachelor of Technology in Computer Science | MAHE, Karnataka, India (2016 - 2020) - CGPA: 8.78/10.0 | Minor Specialisation: Big Data